

Chapter 04: Process Costing

Cost Accounting

What is Process Costing?

Process costing is used most commonly in industries that convert raw materials into homogeneous (i.e., uniform) products, such as bricks, soda, or paper, on a continuous basis.

Similarities between Job-Order and Process

Costing:

Much of what you learned in the previous chapter about costing and cost flows applies equally well to process costing in this chapter. We are not throwing out all that we have learned about costing and starting from “scratch” with a whole new system. The similarities between job-order and process costing can be summarized as follows:

1. Both systems have the same basic purposes—to assign material, labor, and manufacturing overhead costs to products and to provide a mechanism for computing unit product costs.
2. Both systems use the same basic manufacturing accounts, including Manufacturing Overhead, Raw Materials, Work in Process, and Finished Goods.
3. The flow of costs through the manufacturing accounts is basically the same in both systems.

Differences between Job-Order and Process Costing:

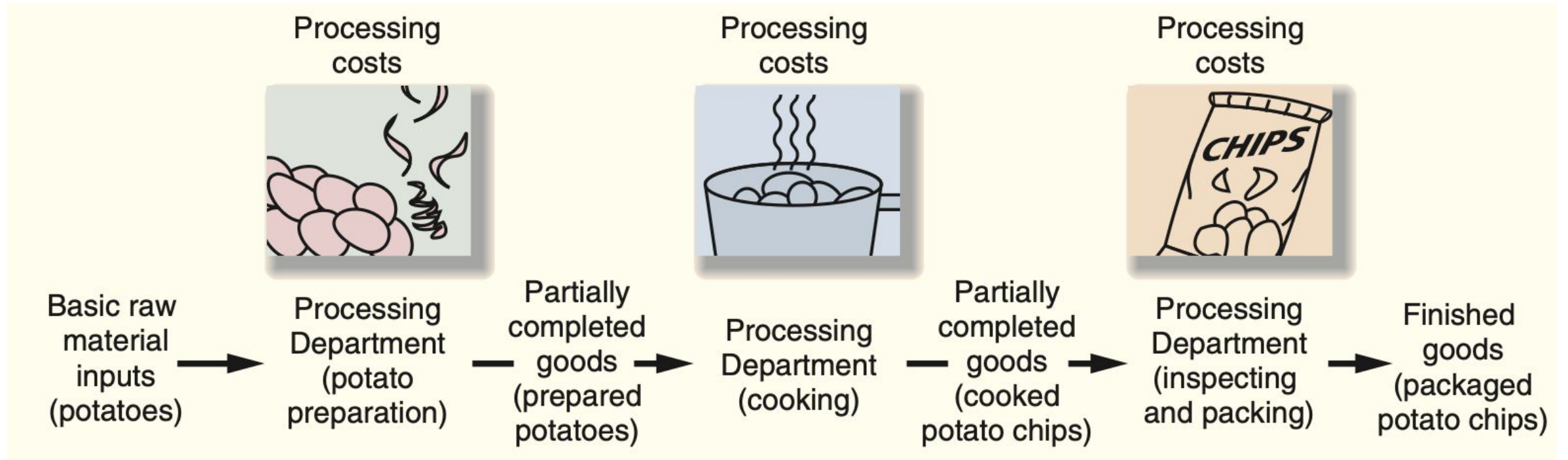
Job-Order Costing

1. Many different jobs are worked on during each period, with each job having different production requirements.
2. Costs are accumulated by individual job.
3. Unit costs are computed *by job* on the job cost sheet.

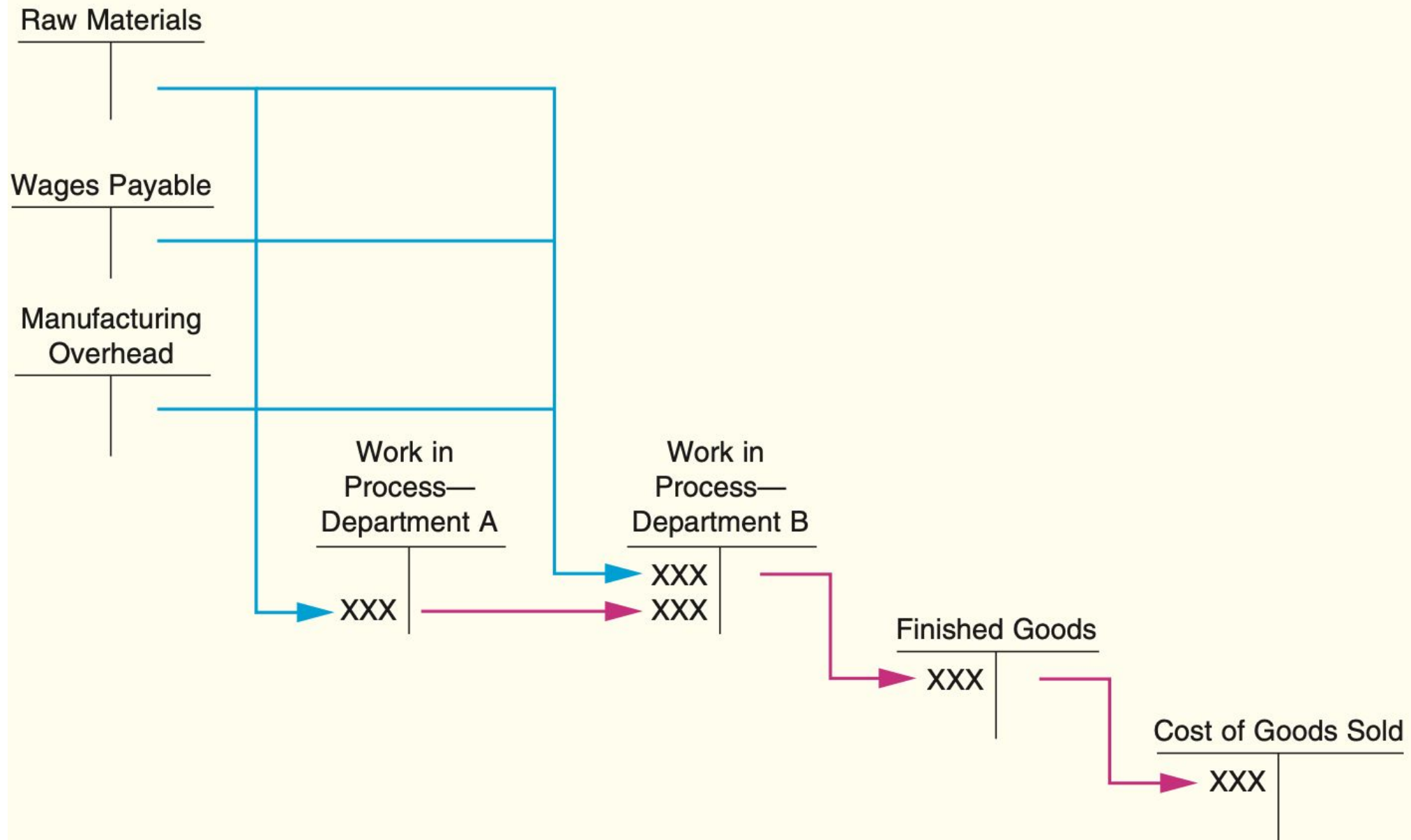
Process Costing

1. A single product is produced either on a continuous basis or for long periods of time. All units of product are identical.
2. Costs are accumulated by department.
3. Unit costs are computed *by department*.

Cost Flows in Process Costing - Processing Departments



Cost Flows in Process Costing Before going through - The Flow of Materials, Labor, and Overhead Costs



Materials, Labor, and Overhead Cost Entries

Materials Costs: As in job-order costing, materials are drawn from the storeroom using a materials requisition form. Materials can be added in any processing department, although it is not unusual for materials to be added only in the first processing department, with subsequent departments adding only labor and overhead costs.

At Megan's Classic Cream Soda, some materials (i.e., water, flavors, sugar, and carbon dioxide) are added in the Formulating Department and some materials (i.e., bottles, caps, and packing materials) are added in the Bottling Department. The journal entry to record the materials used in the first processing department, the Formulating Department, is as follows:

- Work in Process—Formulating Dr XXX
- Raw Materials Cr XXX

The journal entry to record the materials used in the second processing department, the Bottling Department, is as follows:

- Work in Process—Bottling Dr XXX
- Raw Materials Cr XXX

Materials, Labor, and Overhead Cost Entries

Labor Costs: In process costing, labor costs are traced to departments—not to individual jobs. The following journal entry records the labor costs in the Formulating Department at Megan’s Classic Cream Soda:

- Work in Process—Formulating Dr XXX
- Salaries and Wages Payable Cr XXX

A similar entry would be made to record labor costs in the Bottling Department. Overhead Costs In process costing, as in job-order costing, predetermined overhead rates are usually used. Manufacturing overhead cost is applied according to the amount of the allocation base that is incurred in the department. The following journal entry records the overhead cost applied in the Formulating Department:

- Work in Process—Formulating Dr XXX
- Manufacturing Overhead Cr XXX

A similar entry would be made to apply manufacturing overhead costs in the Bottling Department.

Materials, Labor, and Overhead Cost Entries

Completing the Cost Flows Once processing has been completed in a department, the units are transferred to the next department for further processing, as illustrated in the T-accounts in Exhibit 4–3. The following journal entry transfers the cost of partially completed units from the Formulating Department to the Bottling Department:

- Work in Process—Bottling Dr XXX
- Work in Process—Formulating Cr XXX

After processing has been completed in the Bottling Department, the costs of the completed units are transferred to the Finished Goods inventory account:

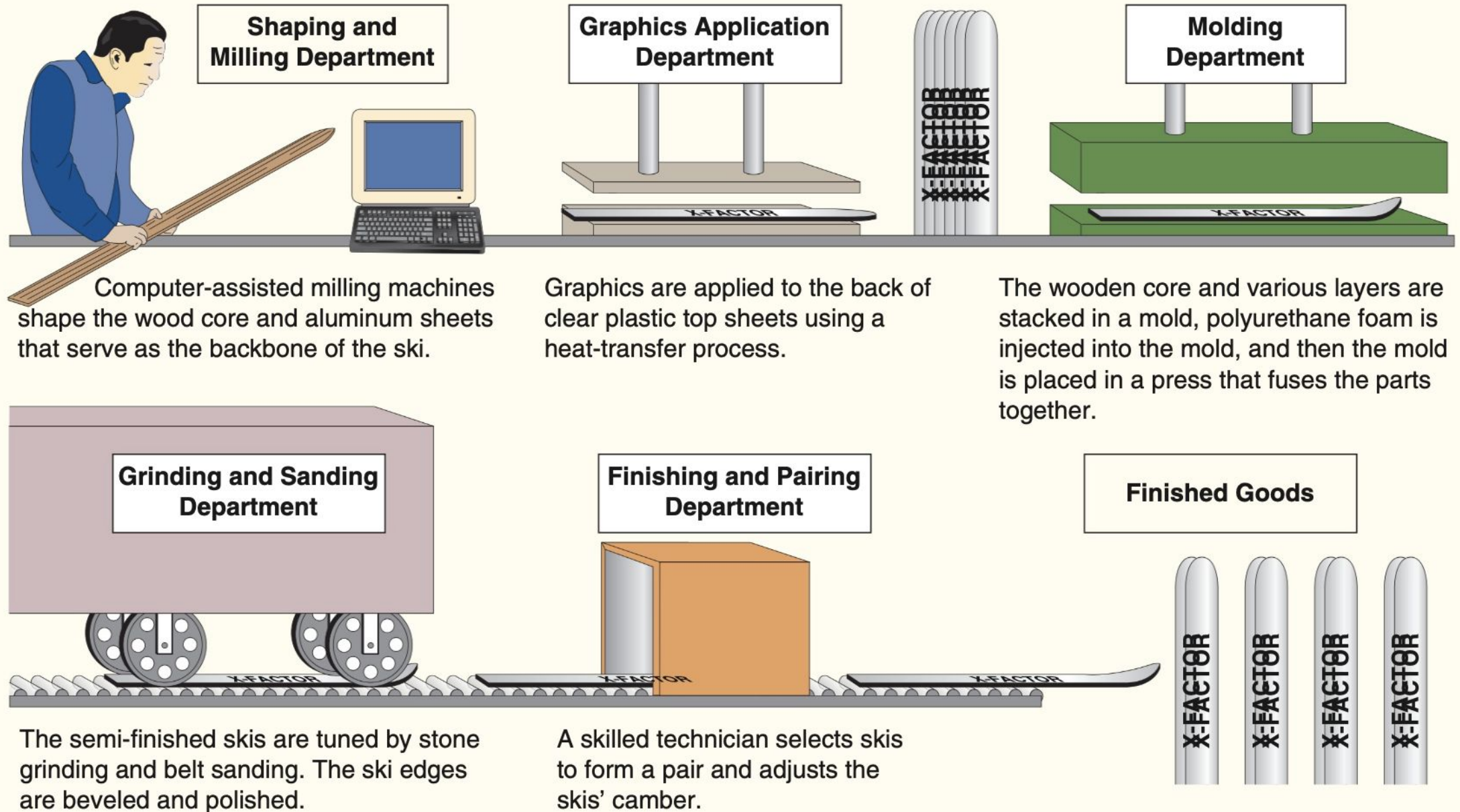
- Finished Goods Dr XXX
- Work in Process—Bottling Cr XXX

Finally, when a customer's order is filled and units are sold, the cost of the units is transferred to Cost of Goods Sold:

- Cost of Goods Sold Dr XXX
- Finished Goods Cr XXX

To summarize, the cost flows between accounts are basically the same in a process costing system as they are in a job-order costing system. The only difference at this point is that in a process costing system each department has a separate Work in Process account.

Equivalent Units of Production



Equivalent Units of Production: Weighted-Average Method

Weighted-Average Method
(a separate calculation is made for each cost category in each processing department)

$$\text{Equivalent units of production} = \text{Units transferred to the next department or to finished goods} + \text{Equivalent units in ending work in process inventory}$$

Suppose the Molding Department at Double Diamond has 500 units in its ending work in process inventory that are 60% complete with respect to processing in the department. These 500 partially complete units are equivalent to 300 fully complete units ($500 * 60\% = 300$). Therefore, the ending work in process inventory contains 300 equivalent units.

Equivalent Units of Production

Consider the Shaping and Milling Department at Double Diamond. This department uses computerized milling machines to precisely shape the wooden core and metal sheets that will be used to form the backbone of the ski. The activity shown at the top of the next page took place in the department in May.

The first thing to note about the activity in the Shaping and Milling Department is the flow of units through the department. The department started with 200 units in beginning work in process inventory. During May, 5,000 units were started into production. This made a total of 5,200 units. Of this total, 4,800 units were completed and transferred to the next department during May and 400 units were still in the department at the end of the month as ending work in process inventory.

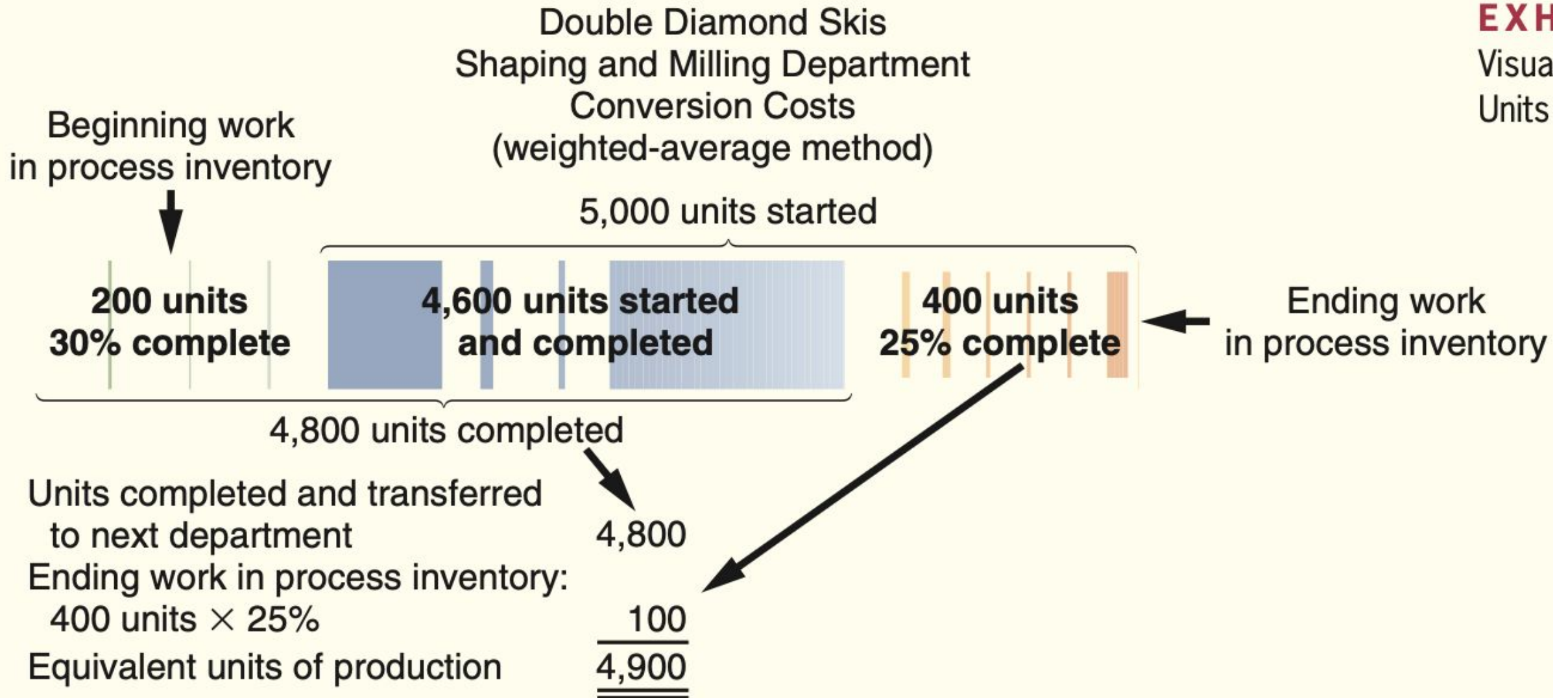
Equivalent Units of Production

Shaping and Milling Department	Units	Percent Complete	
		Materials	Conversion
Beginning work in process inventory	200	55%	30%
Units started into production during May	5,000		
Units completed during May and transferred to the next department	4,800	100%*	100%*
Ending work in process inventory	400	40%	25%
*We always assume that units transferred out of a department are 100% complete with respect to the processing done in that department.			

Equivalent Units of Production

Shaping and Milling Department	Materials	Conversion
Units transferred to the next department	4,800	4,800
Ending work in process inventory:		
Materials: 400 units × 40% complete	160	
Conversion: 400 units × 25% complete	<u> </u>	<u>100</u>
Equivalent units of production	<u><u>4,960</u></u>	<u><u>4,900</u></u>

Equivalent Units of Production



Compute and Apply Costs - Shaping and Milling Department (Data for May Operations)

Beginning work in process inventory:

Units in process	200
Completion with respect to materials	55%
Completion with respect to conversion	30%

Costs in beginning work in process inventory:

Materials cost	\$ 9,600
Conversion cost	<u>5,575</u>
Total cost in beginning work in process inventory	<u><u>\$15,175</u></u>

Units started into production during the period	5,000
Units completed and transferred out	4,800

Costs added during the period:

Materials cost	\$368,600
Conversion cost	<u>350,900</u>
Total cost added during the period	<u><u>\$719,500</u></u>

Ending work in process inventory:

Units in process	400
Completion with respect to materials	40%
Completion with respect to conversion	25%

Cost per Equivalent Unit—Weighted-Average Method

Weighted-Average Method

(a separate calculation is made for each cost category in each processing department)

$$\text{Cost per equivalent unit} = \frac{\text{Cost of beginning work in process inventory} + \text{Cost added during the period}}{\text{Equivalent units of production}}$$

Shaping and Milling Department Costs per Equivalent Unit

	Materials	Conversion
Cost of beginning work in process inventory	\$ 9,600	\$ 5,575
Costs added during the period	<u>368,600</u>	<u>350,900</u>
Total cost (a)	<u><u>\$378,200</u></u>	<u><u>\$356,475</u></u>
Equivalent units of production (see Exhibit 4–5) (b)	4,960	4,900
Cost per equivalent unit (a) ÷ (b)	\$76.25	\$72.75

Applying Costs—Weighted-Average Method

For example, each unit transferred out of Double Diamond's Shaping and Milling Department to the Graphics Application Department, as depicted in Exhibit 4–4, will carry with it a cost of \$149.00 (\$76.25 for materials cost and \$72.75 for conversion cost). Because 4,800 units were transferred out in May to the next department, the total cost assigned to those units would be \$715,200 (4,800 units * \$149.00 per unit).

Applying Costs—Weighted-Average Method

Shaping and Milling Department

Costs of Ending Work in Process Inventory and the Units Transferred Out

	Materials	Conversion	Total
Ending work in process inventory:			
Equivalent units of production (materials:			
400 units $\times$ 40% complete; conversion:			
400 units $\times$ 25% complete) (a)	160	100	
Cost per equivalent unit (see page 154) (b)	\$76.25	\$72.75	
Cost of ending work in process inventory (a) $\times$ (b)	\$12,200	\$7,275	\$19,475
Units completed and transferred out:			
Units transferred to the next department (a) . . .	4,800	4,800	
Cost per equivalent unit (see above) (b)	\$76.25	\$72.75	
Cost of units transferred out (a) $\times$ (b)	\$366,000	\$349,200	\$715,200

Cost Reconciliation Report

Shaping and Milling Department Cost Reconciliation

Costs to be accounted for:

Cost of beginning work in process inventory (Exhibit 4–7)	\$ 15,175
Costs added to production during the period (Exhibit 4–7)	<u>719,500</u>
Total cost to be accounted for	<u><u>\$734,675</u></u>

Costs accounted for as follows:

Cost of ending work in process inventory (see above)	\$ 19,475
Cost of units transferred out (see above)	<u>715,200</u>
Total cost accounted for	<u><u>\$734,675</u></u>